

Topics in this lecture

- Sample space
- Events *เหตุการณ์*
- Probability *ความน่าจะเป็น*

Why Learn Prob & Stat?

- Give you the power to model things that are random in nature
- Can be applied in many fields of study
 - Science
 - Engineering
 - Business

Motivating Example 1

- Netflix grand prize
 - Movie recommendation system
 - Predicts how a viewer will rate a movie based on his/her ratings
- \$1,000,000 prize!!
(Awarded in 2009)



Motivating Example 2

- Performance modeling
 - How many cashiers do I need in order to keep the average waiting time under 5 minutes?



Motivating Example 3

- Classification
 - How does a spam filter know which e-mail is a spam?

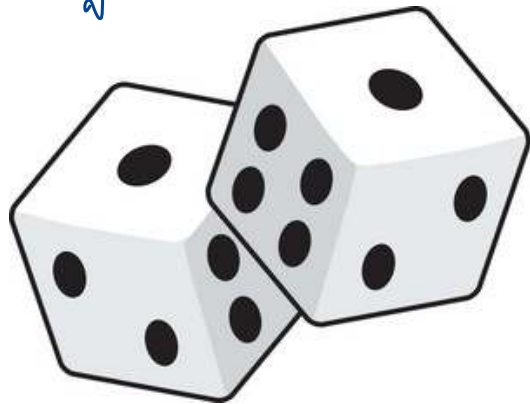


การทดลองสุ่ม

Random Experiments

- Outcome varies in an *unpredictable fashion* when the experiment is repeated under the same conditions
- Must specify clearly what to be measured or observed

โยน 1 ลูก
ผลลัพธ์ที่ได้อาจเป็นอะไรก็ได้



Experiment 1

Select a ball from an urn containing balls numbered 1 to 50. Note the **number of the ball**

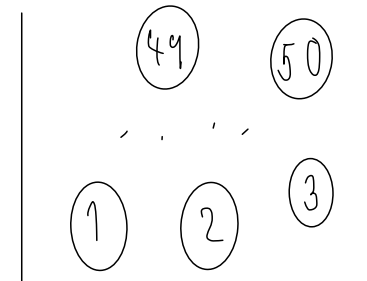
๑. ค่าจะเป็นที่อะไร #บอลเลขที่

$$P(t_1) = \frac{25}{50} = \frac{1}{2} \#$$

Possible outcomes:

1, 2, 3, ...

$$\Omega = \{1, 2, 3, \dots, 50\}$$



Experiment 2

เลือกบอล 1 ลูก

Select a ball from an urn containing balls numbered 1 to 4. Suppose balls 1 and 2 are black and balls 3 and 4 are white. Note the **number and color** of the ball selected.

$$E_2 = \{ (4, W) \}$$

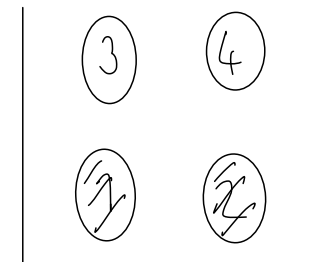
$$P(E_2) = \frac{1}{4} \#$$

๑. ถ้าจะเลือกที่จะได้
บอลที่ ๔ สีขาว

Possible outcomes:

$(3, \text{white}), (2, \text{black})$

$$\Omega = \left\{ \begin{array}{l} (1, B), (2, B) \\ (3, W), (4, W) \end{array} \right\}$$



Experiment 3

โยนเหรียญ 3 ครั้ง

จดลำดับของ H, T ที่ออกหลังจาก 3 ครั้ง

Toss a coin **3 times** and note a **sequence of heads and tails**

$$E_3 = \{ HHH, TTT \}$$

$$P(E_3) = \frac{2}{8} = \frac{1}{4} \neq$$

Possible outcomes:

HHH, THT, ..., TTT

$$\Omega = \left\{ \begin{array}{l} HHH, HHT, HTH, THH \\ TTT, TTH, THT, HTT \end{array} \right\}$$

Experiment 4

โยน 3 เหรียญ

ออกหัวกี่ครั้ง

Toss a coin 3 times and note the number of heads

$$E_4 = \{ \}, \emptyset \quad (\text{โยนไปโยนมา = get nothing})$$

Possible outcomes:

0, 1, 2, 3

$$\Omega = \{0, 1, 2, 3\}$$

$E_{4.1}$ โยน 3 เหรียญ ออกหัว 3 ครั้ง

$$E_{4.1} = \{0, 3\}$$

$$P(E_{4.1}) = \frac{2}{4} = \frac{1}{2}$$

Experiment 5

A packet is transmitted repeatedly over a noisy channel until it is received without error at the receiver. **Count the number of transmissions required.**

ส่งซ้ำๆ จนกว่าจะส่งสำเร็จ



Possible outcomes:

1, 2, 3, ...

$$\Omega = \{1, 2, 3, \dots, 100, \dots\}$$

$$\mathcal{E}_s = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

Experiment 6

สุ่มเลือกตัวเลข 1 ตัว ที่อยู่ระหว่าง 0 กับ 1

Pick a number at random between 0 and 1.

$$E_{6.1} = \text{เลขที่สุ่มได้ 1 ตัว อยู่ระหว่าง 0-1}$$

$$= \{x : 0 < x < 1\}$$

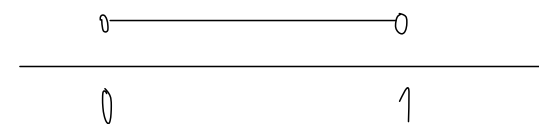
$$P(E_{6.1}) =$$

Possible outcomes:

0.5, 0.1, 0.919, 0.0000000001, ... # เขียนออกมาได้ ๓

$$\Omega_6 = \{0 < x < 1\}$$

เลขทุกตัวไม่ ๐ หรือ ๑ เขียนได้เรื่อยๆ



$$E_6 = \Omega_6$$

Experiment 7

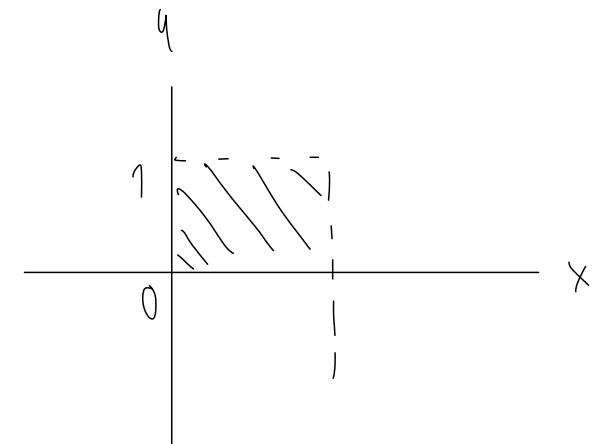
Pick two numbers at random between 0 and 1.

$$\Omega = \{ (x, y) : 0 < x < 1, 0 < y < 1 \}$$

Possible outcomes:

$$(0.5, 0.012), (0.998, 0.123), \dots$$

$$E_7 = \left\{ (x, y) : |x - y| < \frac{1}{10}, \right. \\ \left. 0 < x < 1, 0 < y < 1 \right\}$$



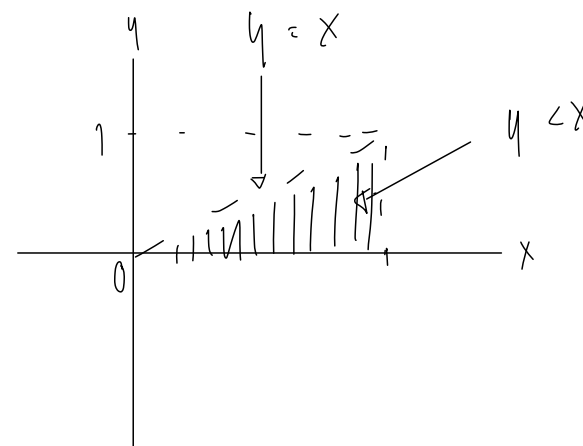
Experiment 8

Pick a number X at random between 0 and 1, and pick a number Y at random between 0 and X .

$$\Omega = \{ (X, Y) : 0 < X < 1, \quad 0 < Y < X \}$$

Possible outcomes:

$$(0.8, 0.2)$$



Sample Space

เปลี่ยน. ทดลอง ทั้งหมดที่ไปไม่ได้
เก็บใน set

- Definition: The sample space S of a random experiment is the *set of all possible outcomes*
- Exercise:
 - Define the sample spaces of Exp1 - Exp8

Types of Sample Spaces

ผลลัพธ์หนึ่ง — นับได้ (แบ่งออกได้)

- **Discrete** — Countable

- Outcome can be put into one-to-one correspondence with positive integers.

- May be finite or countably infinite

นับได้แบบจำกัด , นับได้แบบไม่จำกัด

ต่อเนื่อง

นับไม่ได้ (แบ่งออกไม่ได้)

- **Continuous** — Uncountable

↳ Ex. ช่วงเวลา ที่วิ่งอยู่ระหว่าง 0 ถึง 1

Events

$$\Omega = \{ \dots, \dots \}$$

6 9 0 7 1 1 7 8 2 4 6

← ผลลัพธ์ที่อาจเกิดขึ้น

$$E = \{ \dots, \dots \}$$

↓

- **Subset** of the sample space

- Special events

เหตุการณ์ที่เกิดขึ้นทั้งหมด = sample space Ω

- **Certain event** — the **sample space** itself, consists of all possible outcomes

เหตุการณ์ที่ไม่เกิดขึ้น

- **Impossible event** — the **null event**, contains **no outcome**

$$\emptyset, \{ \}$$

Example Events of Exp1 - Exp8

- E1 = “An even-numbered ball is selected”
ได้บอลเลขคู่
- E2 = “The ball is white and even-numbered”
สีขาวเลขคู่
- E3 = “The three tosses give the same outcome”
โยน 3 ครั้ง ออกมาเหมือนกันหมด
- E4 = “The number of heads equal the number of tails”
จำนวนหัว = etail
- E5 = “Fewer than 10 transmissions are required”
ส่งน้อยกว่า 10 ครั้ง
- E6 = “The selected number is non-negative”
เลขที่เลือก ไม่ติดลบ
- E7 = “The two number differs by less than 1/10”
เลขสองตัวที่ 1 กับ ตัวที่ 2 กัน ไม่เกิน 0.1
- E8 = “The two number differs by less than 1/10”

Experiments Summary

Experiment	Sample Space	Event
1	$S1 =$	$E1 =$
2	$S2 =$	$E2 =$
3	$S3 =$	$E3 =$
4	$S4 =$	$E4 =$
5	$S5 =$	$E5 =$
6	$S6 =$	$E6 =$
7	$S7 =$	$E7 =$
8	$S8 =$	$E8 =$

Creating Events

- *Elementary events* — events from a discrete sample space which consists of a *single outcome*
- Events can be combined using set operations
- Set operations
 - Union
 - Intersection
 - Complement

Set Arithmetic

- Commutative:
 - $A \cup B = B \cup A$
 - $A \cap B = B \cap A$
- Associative:
 - $(A \cup B) \cup C = A \cup (B \cup C)$
 - $(A \cap B) \cap C = A \cap (B \cap C)$

Set Arithmetic (2)

- Distributive:
 - $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
 - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- De Morgan's Rules:
 - $(A \cup B)' = A' \cap B'$
 - $(A \cap B)' = A' \cup B'$

Probability

ความน่าจะเป็น
= ค่าซึ่งที่ กำหนดให้กับผลลัพธ์ของ event

- Probabilities are numbers assigned to events to indicate how “likely” it is that the events will occur
- A probability law is a rule that assigns probabilities to the events of the experiment
- A probability law is a function that *assigns a number to a set (event)*

Simple Probability Assignment

- If the sample space is discrete with n elements and each outcome is *equally likely*, then we can assign each elementary event a probability of $1/n$

ω

หาก n ใด ๆ ใน sample space ต้องมีโอกาสพอ ๆ กัน

$$\frac{1}{n}$$

Example 1.1

- An urn contains 10 identical balls numbered 0, 1, ..., 9. Select a ball and note the number on the ball. Find the probability of the following events:
 - $A = \text{"number of ball selected is odd"}$
 - $B = \text{"number of ball selected is multiple of 3"}$
 - $C = \text{"number of ball selected is less than 5"}$
 - $D = \text{"number of ball selected is odd or less than 5"}$

Example 1.2

- Toss a coin 3 times. Find the probability of getting 2 heads out of 3 tosses.

$$\Omega = \left\{ \begin{array}{l} HHH, HHT, HTH, THH \\ TTT, TTH, THT, HTT \end{array} \right\}$$

$$E_A = \{ HHT, HTH, THH \}$$

$$P(A) = \frac{3}{8} \#$$

நிழலா பரவிய

७७९९ १०११०११

Assigning Probability: Continuous

[illegible]

- Involve assigning numbers to intervals on the real line and region in the plane
- In case of equally likely outcomes, assign probability based on length of interval or size of region in the plane

Q2) Wie viele von 960000 sind

event

sample space

size

Example 1.3

$$\Omega = \{0 < X < 1\}$$

- Pick a random number between 0 and 1. Find the probability of the following events.
- A = “the selected number is less than 0.5” $E_A = \{0 < X < 0.5\}$

$$P(A) = \frac{0.5}{1}$$
- B = “the selected number is between 0.3 and 0.6” $E_B = \{0.3 < X < 0.6\}$

$$P(B) = \frac{0.3}{1}$$
- C = “the selected number is negative” $\emptyset = P(C) = 0$

Example 1.4

$$\Omega = \{ (x, y) : 0 < x, y < 1 \}$$

- Pick 2 numbers x and y at random between 0 and 1. Find the probability of the following events

- $A = "x > 0.5"$ $E_A = \{ 0.5 < x < 1 \}$ $P(A) = \frac{0.5}{1}$

- $B = "y > 0.5"$ $E_B = \{ 0.5 < y < 1 \}$ $P(B) = \frac{0.5}{1}$

- $C = "x > y"$ $E_C = \{ y < x < 1 \}$ $P(C) = 1$

ខ្លឹមសារ ៤ ចំណុច

Axioms of Probability

- Axiom 1:

៖ ប្រូបាប៊ីលីតេ គឺជា លេខ វិជ្ជមាន

$$0 \leq P(A)$$

- Axiom 2:

៖ ប្រូបាប៊ីលីតេ នៃ លំហ គឺស្មើ ១

$$P(\Omega) = 1$$

Axioms of Probability (2)

- Axiom 3:

If $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$

Corollaries

- Corollary 1:

$$P(A') = 1 - P(A)$$

- Corollary 2:

$$P(A) \leq 1$$

- Corollary 3:

$$P(\emptyset) = 0$$

Corollaries (2)

- Corollary 4:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$